

# Dream bigger, feel worse? The side effects of boosting aspirations

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## Abstract

We study the impact of a brief, low-cost intervention on aspirations, soft skills and well-being. 297 primary and middle school students from fragile backgrounds in Naples, Italy, were randomized into viewing a short optimistic video message from a role model. We find that exposure to the video significantly increases short-term self-reported educational aspirations, especially among immigrant students. However, it causes a shift in self-efficacy and grit, a decline in mood, and no measurable improvement in long-term academic performance. This study shows how a scalable role model intervention impacting short-run aspirations creates backlashes on students' well-being and attitudes.

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# 1 Introduction

Low aspirations among children from disadvantaged backgrounds have long been recognized as a barrier to human capital accumulation and upward mobility. Theoretical work highlights aspirations as a key determinant of educational investment and, ultimately, labor market outcomes (Genicot and Ray (2017)). As people from low socio-economic background tend to have lower and more fragile means to aspire (money, social network, mentoring, etc.), poverty is a cause of lower aspirations. At the same time, as low aspiration leads to lower investments in education and in turn lower wages, poverty is also an effect of lower aspirations. This phenomenon has been addressed in the literature as "aspiration trap" (Appadurai (2004); Banerjee et al. (2006); La Ferrara (2019)), a self-reinforcing, vicious mechanism that condemns poor individuals to poverty.

Among others<sup>1</sup>, interventions featuring positive role models have been found effective in raising student's aspirations and, eventually, their academic achievements, even in the long run (Riley et al. (2017); Porter and Serra (2020)). However, there is little evidence on the impact of such intervention on other student's outcomes, like attitudes (self-efficacy, grit) and mood. This is surprising and constitutes a gap in the literature, given the correlation of these factors with effort and, therefore, success (Alan et al., 2019; Correa-Rojas et al., 2024).

This paper aims to shed light on the unintended consequences of a role model intervention, designed to boost aspiration, on attitudes and well-being. We survey 297 primary and middle school students in Naples (Italy), and randomly assign them to watch a one-minute video delivering a message of hope. Our sample is composed by a greater share of children from low SES and immigrant background<sup>2</sup> compared to the median Italian school. As these students

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<sup>1</sup>Several interventions have been tested and implemented to break this vicious circle and increase aspirations among students, from organizing tutoring programs and academic counseling (Carlana et al. (2022)) to institutional changes in the political rules (Beaman et al. (2012)) and the provision of statistical information on the benefits of investing in education (Nguyen (2008)).

<sup>2</sup>In this paper, we define having an immigrant background as being born outside of Italy or having at least one parent born outside of Italy, irrespectively of nationality. Conversely, Italian students are defined as students whose parents are both born in Italy.

are the most exposed to the risk of under-aspiring (Carlana et al. (2022)), they are the main target of our intervention. For this reason, and following the literature stating that role models from more difficult background positively impact all students (Olsson and Martiny (2018)), we recruited actors who had an immigrant background.

We find that all treated students report significantly higher aspirations, but this backfires on a composite index of grit, self-efficacy, general well-being, and mood, differentially based on immigrant status. The video lowers Italian students' grit and increases their self-efficacy, possibly as it pushes them to think of success as more achievable. On the contrary, the treatment decreases immigrants' self-efficacy, possibly due to increased salience of their future choices. Both groups report a negative impact on mood. This negative effect might be explained by the intentional vagueness of the treatment, which does not inform students on steps to take in order to pursue their goals, but simply conveys a positive message of hope. This lack of information, which is reflected also on qualitative evidence collected during the interviews, might spark a feeling of uncertainty in the future and translate into anxiety. Future research should shed light on these mechanisms.

The theory regards the potential impact of role model interventions on attitudes and well-being is ambiguous. On the one side, hope (which we *hoped* to convey through our role model intervention) has been found to be positively related to self efficacy (Magaletta and Oliver, 1999) and well-being (Murphy, 2023; Pleeging et al., 2021). On the other side, according to Bandura (1986) when individuals think about an uncertain or ambiguous future, they can feel less in control over outcomes and this might decrease their grit, or they can become more anxious about the future, reducing their perceived efficacy. This process can ultimately affect their self worth and, in turn, their mood and well-being (Bandura (1997); Bandura et al. (2003); Siddiqui (2015) . Additional explanations for the decrease in mood follow Anderson et al. (2019), which links increased uncertainty (such as the one potentially arising from observing a hopeful video

that pushes the student to think about the future without giving any actionable information) to a worsening in affect - which we measure through our "mood" scale.

This paper makes three contributions to the literature on aspirations, human capital formation, and labor economics. First, it shows that a brief, cost-effective motivational video that lasts approximately 1 minute can indeed boost short-term aspirations. This very short and at the same time virtual intervention is a novelty in the literature, where pre-recorded interventions are in the form of movies or TV shows, and therefore last longer (Bjorvatn et al. (2020); Riley et al. (2017)). Shorter interventions are instead a feature of some in-person events with real role models taking the stage and interacting with the sample (Nguyen (2008)). Despite not persisting in the long term, the short term impacts we find might be important in contexts where pivotal choices exist and are meant to be taken in specific moments, such as high school choices in Italy or enrollment in college courses. Second, it collects a variety of information not only on student's school and aspirations, but also on student's grit, self-efficacy, mood and well-being, to assess the impact - and possible side effects - of the intervention on such outcomes. Third, contrary to most papers in the literature which are run in the Global South, this experiment is conducted in a school in Naples, Italy. It therefore allows to analyse the effects of role models in a high income country on a sample of primary and middle school students from fragile socio-economic backgrounds.

## 2 Context

What follows is based on data collected by means of in-person interviews<sup>3</sup> in a primary and middle school in the center of Naples, attended by a high share of students from low socio-economic and immigrant background. The sample is balanced in terms of gender composition, with 150 boys and 145 girls. This figure is balanced also when separately looking at primary and

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<sup>3</sup>The interviews were conducted in the month of May 2021 by a team of 9 enumerators.

middle school subgroups. 59% of the sample is made of Italians, 25% have instead a foreign nationality. Overall, 33% reports at least one of their parents having a foreign nationality. Henceforth, we refer to this subsample who is born abroad or has at least a parent born abroad as "Second generation Immigrants". We remand to Table 1 for sample characteristics.

We measure aspirations as self-reported choice of high school. In the Italian schooling systems, pupils attend 8 years of compulsory universal schooling (G1-G5 in elementary school, G6-G8 in middle school). At the end of middle school, they choose which high school to attend, based on their experience in the schooling system so far and on their aspirations for the future. Most high schools are public and free to attend and do not require entry examinations. High schools differ primarily in terms of topics covered and of certifications released at the end of their 5 year duration: while the vocational and technical tracks deliver a certification that may allow students to enter the job market, the more ambitious academic track does not, and is usually followed by enrollment in University (Carlana et al., 2022).

Overall, many children in our sample have low aspirations compared to the rest of the country. While in Italy up to 56% of pupils choose to follow an "academic" high school in 2025 (MIM, 2025), in our control sample native students intend to choose the "academic" high school with a frequency of 44%, while this is chosen by immigrants in 39% of cases. As additional evidence for low aspirations, we provide the self-reported job aspirations of the control group. As illustrated by the word cloud represented in Figure A1, the most cited jobs are hairdresser, beautician, and doctor for girls, and soccer player, doctor, and cook for boys.

### **3 Intervention**

The intervention consists in showing a short motivational video to a randomly selected sample of students. The treatment group was shown a 1-minute video in which an actor or an actress

Table 1: Balance table

|                                     | Means / sd |        | Differences / se | P-values |        | Obs |
|-------------------------------------|------------|--------|------------------|----------|--------|-----|
|                                     | C          | T      |                  | T-C      | Simple |     |
| <b>Panel A. Demographics</b>        |            |        |                  |          |        |     |
| Primary                             | 0.86       | 0.77   | -0.10            | 0.04     | 0.95   | 297 |
|                                     | 0.35       | 0.42   | 0.05             | .        | .      | .   |
| Female                              | 0.56       | 0.49   | -0.07            | 0.24     | 0.97   | 297 |
|                                     | 0.50       | 0.50   | 0.06             | .        | .      | .   |
| Immigrant 2nd gen                   | 0.24       | 0.28   | 0.04             | 0.52     | 0.58   | 297 |
|                                     | 0.43       | 0.45   | 0.05             | .        | .      | .   |
| Immigrant 2nd gen missing           | 0.16       | 0.15   | -0.01            | 0.84     | 0.48   | 297 |
|                                     | 0.37       | 0.36   | 0.05             | .        | .      | .   |
| Single parent                       | 0.32       | 0.21   | -0.11            | 0.05     | 0.89   | 297 |
|                                     | 0.47       | 0.41   | 0.06             | .        | .      | .   |
| <b>Panel B. Administrative data</b> |            |        |                  |          |        |     |
| Absences 20/21 (std.)               | -0.20      | -0.065 | 0.18             | 0.14     | 0.66   | 289 |
|                                     | 1.00       | 0.97   | 0.12             | .        | .      | .   |
| Absences 20/21 missing              | 0.032      | 0.025  | -0.01            | 0.75     | 0.82   | 297 |
|                                     | 0.18       | 0.16   | 0.02             | .        | .      | .   |
| Italian 20/21 (std.)                | 0.19       | 0.048  | -0.14            | 0.29     | 0.91   | 273 |
|                                     | 1.04       | 0.97   | 0.13             | .        | .      | .   |
| Math 20/21 (std.)                   | 0.23       | 0.042  | -0.19            | 0.18     | 0.95   | 273 |
|                                     | 1.09       | 0.94   | 0.14             | .        | .      | .   |
| Joint balance test                  | .          | .      | .                | 0.10     | 1.00   | .   |

We report standard deviations in columns 1, 2 and robust standard errors in columns 3, below averages. Primary school: indicator for the student being in primary school (=0 if student is in middle school). Female: indicator for female (=0 if male). Immigrant 2nd gen: indicator for having at least a parent born abroad, as reported by the student. Single parent: the child reports living only with one parent. Absences: hours the student was expected to be in school but was not (for any reason), standardized from administrative data separately for middle and elementary school. Grades: yearly average grade in Italian or Maths, standardized separately for elementary and middle schools. Absences and Grades refers to school year 2020/2021, only if students were enrolled in the same school the previous year. Differences in number of hours absent and in grades are reported controlling for class fixed effects. Joint balance tests: *p-value* from the F-test of regressing treatment assignment on the controls, including flags when controls are missing. We report P-values for simple differences and for differences weighted using entropy weights computed by program ebalance ((Hainmueller and Xu, 2011)) using, as balancing variables, absences, marks in Italian and Maths and a dummy for missing information on administrative data in the previous year, a dummy for gender, immigrant status and single parent.

tells the story of how, starting from a problematic situation during primary and middle school, they found their passion at some point in their life, and this was the key to their current happiness. The purpose of the video treatment is to inspire hope and confidence in the viewer, by instilling in them the idea that even if their situation in school and family is hard and they feel they do not fit, they can still be happy in life by pursuing their passions. Importantly, the purpose of the video was not to boost interest in any specific high school track.

Actors were people living in Naples but originally – and visibly – of a nationality other than Italian, aged between 20 and 30 years old, and both speaking in a Neapolitan accent. The reason we chose actors with these characteristics was to increase the chances the respondent would relate to them, considering that a large portion of the sample are first- or second-generation immigrants. In the treatment video, the story does not include the specific steps the actor took to find their passions and achieve their current happiness: this was done intentionally, in order not to link this message of hope to one specific set of life choices, and to allow as many students as possible to find similarity with the video. The control group was shown a video of the same actor or actress thanking the student for participating in the data collection<sup>4</sup>.

When asked, 95% of students reported having understood and 85% reported having paid attention to both videos, albeit they reported liking the treatment video less (-12 p.p.  $p - value = 0.008$ ). These results lead us to believe the video was followed and understood by most of the students.

The videos were shown during the first and only round of interviews, minutes before our measures of aspirations, self-efficacy, grit, well-being and mood were collected. This implies that this RCT does not envisage the collection of baseline data by design, and that the identification relies on the randomization between treatment arms. The video was shown to the respondent,

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<sup>4</sup>We recorded four videos in total: one treatment and one control video for each of the two actors, female and male. Videos were self-recorded in a very spontaneous manner with the internal camera of a cell-phone, and the only identifiable element of the video was the face of the actors. See Appendix 2 for the scripts and pictures of the videos.

on the tablet given to the enumerator, during the interview, after the first set of questions meant to collect demographics and background data of the respondent.

Treatment assignment is stratified by grade – primary and middle school – and is performed at the individual level with a computer program, several days prior to the intervention. In particular,  $2/3$  of the students were assigned to view the treatment video, and  $1/3$  to view the control video. We remand to Table 1 for balance checks between treatment arms. Since the treatment assignment has resulted unbalanced on 2 out of 9 baseline variables and the Joint balance test is marginally significant, we also provide estimates of impact using entropy balancing weights computed using the Stata program ebalance from Hainmueller and Xu (2011). Table 1 also reports balance checks using such weights.

## 4 Methodology

### Data

The outcomes of interest were measured through in-person surveys in May 2021, straight after the viewing of the video, and consist of self-reported school aspirations, grit, self-confidence and well-being. Additionally, we collected administrative data on students’ grade and absences in the year following the intervention (school year 2021/22).

A first category of outcomes is related to the aspirations of the students. The following open question was asked during the interview: “Which type of high school, if any, would you like to attend when you grow up?” The answers were then coded into five categories: Professional (3 or 5 year schools teaching a set of skills directly implementable in the workplace), Technical (5 years of high school which can give access to university but also directly employable after the end of the studies), Academic (or “Liceo” in Italian, a track which does not teach skills directly employable in jobs but is designed to guide students to University studies), No school

(for those students who did not plan to enroll in any high school<sup>5</sup>), and Don't know, for those students who have not decided yet.

A second category of outcomes that might be affected by the message of the video are measures of grit, self-efficacy, well-being and mood. These self-reported outcomes are measured with four distinct groups of questions (we refer to Online Appendix A for the list of questions included in the four indicators). All questions asked had a 4-level agreement scale, on which the respondent could locate themselves. The indexes were then constructed with Principal Component Analysis (PCA). Finally, the four constructed indexes were averaged out to summarize the students' characteristics and assess the impact of the treatment on their attitudes.

A third category of outcomes is the administrative data of students in the subsequent school year, which were given by the school itself. For this reason, students in grade 8th at the time of the intervention were not included anymore in the sample analysed, as well as other students who switched schools between the years. We report impact on grades given by teachers in the subjects of Italian, Math and English. Additionally, we compute the Grade Point Average (GPA) as the average of all subjects taught in each class and the total hours the pupil was absent. We report results on these measures standardized using the mean and standard deviation of the control group, separately by elementary or middle school.

## Estimation

We assess the impact of treatment by running a simple difference analysis represented by the following regression:

$$y_{ic} = \alpha_c + \beta \text{Treat}_{ic} + \delta X_{ic} + \varepsilon_{ic} \quad (1)$$

where  $y_{ic}$  is the outcome of interest for student  $i$  in class  $c$ ,  $\text{Treat}_{ic}$  is a dummy that assumes value 1 if the student watched the treatment video and 0 if they watched the control;  $X_{ic}$  is

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<sup>5</sup>Although schooling is compulsory until 16, students could report not wanting to enroll in any high school in our questionnaire

a vector of controls, i.e. gender of the respondent, a dummy for whether the student reports being first or second generation immigrant, a dummy for whether their immigrant status is missing, an indicator for single parent.  $\alpha_c$  are class fixed effects. We also separately assess the impacts on the population of Italian students or self-reported immigrants, by interacting the treatment indicator with  $Immigrant_{ic}$ , an indicator equal to 1 if the child is born abroad or reports having at least one parent born abroad, as in the following regression:

$$y_{ic} = \alpha_c + \beta_1 Italian_{ic} * Treat_{ic} + \beta_2 Immigrant_{ic} * Treat_{ic} + \delta X_{ic} + \varepsilon_{ic} \quad (2)$$

As stated above, given the imbalance in the treatment assignment, we report as main outcomes the treatment impact computed using the entropy weights as calculated by the ebalance program (Hainmueller and Xu, 2011), using as balancing variables the ones which were found to be significantly different among the treatment and the control group: absences and marks in Italian and Maths in the two previous years, and an indicator for the child living with an only parent. For the heterogeneity of treatment between immigrants and italians, we compute weights separately on each subsample and combine them in one variable. We report robust p-values in parentheses and p-values from Fisher-permutations test based on 1,000 permutations of the treatment assignment in brackets.

All attitudes indexes and administrative school outcomes are standardized in order to obtain an average of 0 and a standard deviation of 1 in the control sample in the regression. In our main analysis, since we use entropy weights, we perform the standardization using the same entropy weights which we use in the main regression. Online Appendix A reports the same estimations computed on the unweighted sample on outcomes standardized without weights.

## 5 Results

Table 2 reports the overall effects of viewing the treatment video on students' aspirations, and the differential effects on Italian and on first or second generation immigrants. We find that the intervention boosts self-reported aspirations for academic high-school for all treated students, as it increases the number of immigrant children who would like to enroll in an academic track by 33 p.p. (from a base value of 32% in the weighted control sample) and increases the choice of the academic track among Italian children by 16 p.p. (from a base value of 39%). The treatment also causes a decline of the percentage of immigrant children who would like to pursue technical education (-25 p.p., from a base value of 26%) and of those who do not know yet (-10 p.p. from a base value of 28%, however not significant at 10% confidence level).

Table 2: Treatment impact on School aspirations

|               | Academic                 |                          | Technical                 |                           | Professional              |                           | No school                 |                           | Don't know                |                           |
|---------------|--------------------------|--------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|
|               | (1)                      | (2)                      | (3)                       | (4)                       | (5)                       | (6)                       | (7)                       | (8)                       | (9)                       | (10)                      |
| Treat         | 0.15<br>(0.01)<br>[0.05] |                          | -0.05<br>(0.23)<br>[0.23] |                           | -0.07<br>(0.13)<br>[0.19] |                           | -0.01<br>(0.64)<br>[0.70] |                           | -0.06<br>(0.33)<br>[0.38] |                           |
| ItalianXT     |                          | 0.16<br>(0.04)<br>[0.06] |                           | 0.00<br>(0.92)<br>[0.93]  |                           | -0.06<br>(0.29)<br>[0.27] |                           | -0.03<br>(0.48)<br>[0.45] |                           | -0.11<br>(0.18)<br>[0.12] |
| ImmigrantXT   |                          | 0.33<br>(0.00)<br>[0.00] |                           | -0.25<br>(0.04)<br>[0.00] |                           | -0.05<br>(0.41)<br>[0.41] |                           | 0.02<br>(0.50)<br>[0.61]  |                           | -0.10<br>(0.47)<br>[0.28] |
| Mean Control  | 0.37                     |                          | 0.13                      |                           | 0.17                      |                           | 0.06                      |                           | 0.27                      |                           |
| MC Italians   |                          | 0.39                     |                           | 0.08                      |                           | 0.18                      |                           | 0.09                      |                           | 0.27                      |
| MC Immigrants |                          | 0.32                     |                           | 0.26                      |                           | 0.14                      |                           | 0.00                      |                           | 0.28                      |
| R2            | 0.16                     | 0.16                     | 0.18                      | 0.21                      | 0.16                      | 0.19                      | 0.09                      | 0.10                      | 0.12                      | 0.12                      |
| Obs.          | 297                      | 297                      | 297                       | 297                       | 297                       | 297                       | 297                       | 297                       | 297                       | 297                       |

P-values from robust standard errors in parenthesis; p-values from randomized inference with 1000 repetitions in brackets. Includes classroom fixed effects. Sample weighted using entropy weights computed by program ebalance ((Hainmueller and Xu, 2011)) using, as balancing variables, absences and marks in italian and math in the two previous years and only parent. Outcomes are dummies for self reported high school choice.

Beyond boosting aspirations, we find that the treatment negatively impacts attitudes for both Italian and immigrant students, however displaying very different impacts on the two groups. In particular, it decreases reported self-efficacy for immigrant students (-0.68 sd of the self-

Table 3: Treatment impact on attitudes and wellbeing

|               | Grit   |        | Self-Efficacy |        | Wellbeing |        | Mood   |        | Index  |        |
|---------------|--------|--------|---------------|--------|-----------|--------|--------|--------|--------|--------|
|               | (1)    | (2)    | (3)           | (4)    | (5)       | (6)    | (7)    | (8)    | (9)    | (10)   |
| Treat         | -0.30  |        | 0.06          |        | 0.04      |        | -0.60  |        | -0.22  |        |
|               | (0.07) |        | (0.67)        |        | (0.84)    |        | (0.01) |        | (0.03) |        |
|               | [0.14] |        | [0.76]        |        | [0.87]    |        | [0.03] |        | [0.07] |        |
| ItalianXT     |        | -0.54  |               | 0.31   |           | 0.09   |        | -0.66  |        | -0.19  |
|               |        | (0.00) |               | (0.06) |           | (0.66) |        | (0.02) |        | (0.10) |
|               |        | [0.00] |               | [0.07] |           | [0.69] |        | [0.01] |        | [0.10] |
| ImmigrantXT   |        | 0.46   |               | -0.68  |           | -0.12  |        | -0.94  |        | -0.44  |
|               |        | (0.21) |               | (0.02) |           | (0.80) |        | (0.04) |        | (0.08) |
|               |        | [0.06] |               | [0.00] |           | [0.69] |        | [0.01] |        | [0.01] |
| Mean Control  | 0.00   |        | 0.00          |        | 0.00      |        | 0.00   |        | 0.02   |        |
| MC Italians   |        | 0.07   |               | -0.23  |           | -0.02  |        | 0.09   |        | -0.03  |
| MC Immigrants |        | -0.18  |               | 0.59   |           | 0.05   |        | -0.26  |        | 0.14   |
| R2            | 0.13   | 0.15   | 0.11          | 0.12   | 0.14      | 0.13   | 0.12   | 0.12   | 0.11   | 0.11   |
| Obs.          | 283    | 283    | 290           | 290    | 276       | 276    | 268    | 268    | 292    | 292    |

P-values from robust standard errors in parenthesis; p-values from randomized inference with 1000 repetitions in brackets. We control for gender, immigrant status, a dummy for missing information on immigrant status, single parent. Includes classroom fixed effects. Sample weighted using entropy weights computed by program ebalance ((Hainmueller and Xu, 2011)) using, as balancing variables, absences, marks in Italian and Maths and a dummy for missing information on administrative data in the previous year, a dummy for gender, immigrant status and single parent. Outcomes are standardized grit, self efficacy, wellbeing and mood measures constructed with Principal Component Analysis from items listed in Online Appendix 1 using the balancing weights, and a comprehensive index averaging the four.

efficacy index), displaying however a positive impact on their grit. Treated native students, instead, show an increase in self-efficacy (+0.31 sd), but a decrease in grit (-0.54 sd). Both groups report a backlash on the mood indicator (-0.94 sd for immigrant students and -0.66 sd for Italian students).<sup>6</sup>

While the intervention shows encouraging short-term effects on students' aspirations, we find no evidence of long-term impacts on educational outcomes. Using administrative data from the 2021/22 school year—specifically grades and school absences recorded one year after the intervention—we observe a series of null results, as reported in Table 4. We attribute this lack

<sup>6</sup>Even though among immigrants the treatment modifies grit or self efficacy, it does not seem that this mediates the increase in aspirations. Indeed, the indirect effect is zero and the direct effect is as large as the total effect among both italians and immigrants. Separate mediation analysis for italians and immigrants suggest that immigrants' likelihood of reporting a preference for a high school track is positively affected by their increase in grit when viewing the video (10% or 4 p.p. of the total effect is indirect), while the other components of the index do not have a significant impact in determining the decisions. For italians, the index components do not mediate the impact of the video.

Table 4: Treatment impact on School outcomes

|               | Italian                   |                           | English                   |                           | Maths                    |                          | GPA                       |                           | Absences                 |                          |
|---------------|---------------------------|---------------------------|---------------------------|---------------------------|--------------------------|--------------------------|---------------------------|---------------------------|--------------------------|--------------------------|
|               | (1)                       | (2)                       | (3)                       | (4)                       | (5)                      | (6)                      | (7)                       | (8)                       | (9)                      | (10)                     |
| Treat         | -0.01<br>(0.95)<br>[0.95] |                           | -0.12<br>(0.43)<br>[0.49] |                           | 0.11<br>(0.47)<br>[0.55] |                          | -0.08<br>(0.61)<br>[0.64] |                           | 0.20<br>(0.13)<br>[0.15] |                          |
| ItalianXT     |                           | -0.03<br>(0.88)<br>[0.89] |                           | -0.06<br>(0.73)<br>[0.76] |                          | 0.14<br>(0.38)<br>[0.41] |                           | -0.06<br>(0.73)<br>[0.71] |                          | 0.14<br>(0.34)<br>[0.33] |
| ImmigrantXT   |                           | 0.22<br>(0.51)<br>[0.34]  |                           | -0.07<br>(0.87)<br>[0.80] |                          | 0.24<br>(0.46)<br>[0.35] |                           | 0.09<br>(0.80)<br>[0.71]  |                          | 0.31<br>(0.31)<br>[0.09] |
| Mean Control  | 0.00                      |                           | -0.00                     |                           | 0.00                     |                          | -0.00                     |                           | 0.00                     |                          |
| MC Italians   |                           | -0.02                     |                           | -0.08                     |                          | 0.00                     |                           | -0.03                     |                          | 0.11                     |
| MC Immigrants |                           | 0.06                      |                           | 0.24                      |                          | -0.01                    |                           | 0.08                      |                          | -0.33                    |
| R2            | 0.11                      | 0.10                      | 0.12                      | 0.13                      | 0.14                     | 0.14                     | 0.11                      | 0.10                      | 0.22                     | 0.22                     |
| Obs.          | 248                       | 248                       | 247                       | 247                       | 248                      | 248                      | 248                       | 248                       | 248                      | 248                      |

P-values from robust standard errors in parenthesis; p-values from randomized inference with 1000 repetitions in brackets. Includes classroom fixed effects. Sample weighted using entropy weights computed by program ebalance ((Hainmueller and Xu, 2011)) using, as balancing variables, absences and marks in italian and math in the two previous years and only parent. Outcomes are standardized administrative grades, GPA computed by the researchers and absences from school year 2021/22.

of persistence to the nature of the intervention: its brevity and light-touch design may have been insufficient to influence sustained academic effort, particularly in the context of the many competing stimuli students are exposed to throughout a school year.

We believe this paper presents two important results. First, the intervention shows that aspirations can be boosted by short and inexpensive videos, that might be used right before *pivotal moments*, when students make important choices about their future. In our context, we target the choice of high school - which in Italy is performed by students right after the end of grade 8th, and determines the education and the range of jobs the person will have access to in the future. In different contexts, similar moments may occur at the end of high school, when choosing university majors, or when applying for first internships: low aspirations at this point may hamper career trajectories of students from fragile background. In fact, a common problem for individuals with low aspirations is their reluctance to accept challenges that they think they will not be able to overcome, and we show that a short and inexpensive video could

encourage participants to enroll in higher track schools, thus placing them, only in virtue of their choice, on a higher income trajectory.

Conversely, our study cautions on the side effects of interventions aiming to boost aspirations. All students in the treatment group experience a decrease in our attitude and mood index, thus showing that projecting their thoughts in the future, even with a hopeful message, may introduce anxiety or make them uncertain about the future. As the attitudes we measure have been shown to predict educational outcomes, it is crucial to ensure that such interventions do not, in attempting to give with one hand, take away with the other.

The results are even more interesting in light of the different reactions in the minority and majority students: minority students (immigrants) see their grit increase (not significantly) and their self efficacy decrease. This result might signal that the treatment made them aware of their limits, and therefore the cost of increasing their aspiration might be an increase in their perceived probability of failure, which may explain the drop in their well-being. On the other side, majority students (Italians) see a decrease in their grit and an increase in their self efficacy, possibly boosted by the acknowledgment of success of someone from a more difficult background. Still, thinking about the future causes for both groups a decrease in mood compatible with an increase in anxiety. In sum, our work points at the importance, when evaluating role model interventions, to assess their impact on students' attitudes and well being as well.

## 6 Conclusions

Can interventions aimed at raising aspirations have unintended downsides? Our findings suggest that they can. While our light-touch, virtual role model intervention successfully elevated students' aspirations, it simultaneously led to declines in mood and self-reported attitudes. This adverse effect may stem from the video prompting feelings of uncertainty about the future or performance-related anxiety—observed among both immigrant and Italian students.

We find a significant effects on school aspirations, which are overall higher among the treated students, in a comparable way as the literature Porter and Serra (2020). Why the negative effects on mood and attitudes? Both qualitative and quantitative data collected during the interviews point to the lack information children have to make choices, which might trigger their negative perception of the future. Future research may focus on identifying the mechanisms underlying this effect.

We find no effects of our light touch intervention on future grades and school outcomes: we reconcile this findings with previous literature by acknowledging the limited scope of our intervention. Riley et al. (2017) finds an effect on exam performance because the long movie in her intervention showed a character with a positive academic behaviour at a time when students were focused on improving their academic standing by studying for the national standardized exam. In our case the video was only 1 minute long, the actors never mention effort in academic setting, and during the course of the next academic year (when outcomes were measured) the students will have received a huge number of similar stimuli.

The significance of our results relies on the extent to which children’s aspirations and confidence can easily be boosted in the short run by a short and inexpensive intervention, albeit only in the short run. However, we posit that the short-run effect could be sufficient to create a long-run change in one’s academic and career path if the intervention is administered prior to taking sensitive decisions, such as school choice or participating in school activities which could improve one’s curriculum or skills. Further research is needed to understand if such choices would be beneficial for the student, and position them on a permanent ”better” path, or if the students would be harmed by choosing options ”out of their league”.

Our findings highlight the importance of considering potential adverse effects on well-being when designing interventions aimed at fostering ambition. Furthermore, we observe heterogeneous impacts on grit and self efficacy across student groups, depending on their perceived

similarity to the role model and their assessment of the model's background. Specifically, Italian students, who believe they face fewer challenges than the role model, tend to exhibit reduced grit, possibly because they perceive success as more easily attainable. In contrast, immigrant students, who may identify more closely with the hardships portrayed in the video, show a potential increase in grit but a decrease in self-efficacy, as the narrative may prompt reflection on their own limitations.

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## Online Appendix A

Figure A1: Wordcloud of job aspirations in the control group.



Table A1: Item-Level Measures for Index Construction

| Index         | Item                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Grit          | 1. If I think I can lose at a game, I don't want to keep playing<br>2. When I receive a bad result on a test, I spend less time on this subject and focus on other subjects that I'm actually good at<br>3. I prefer hardest homework that makes me reflect a lot, rather than easy homework that I can complete easily                                                                                                                                                                            |
| Self-efficacy | 1. I can do things independently<br>2. If I have a problem, I can fix it                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Wellbeing     | 1. Do you think you can do whatever you want in your life?<br>2. Are you happy in general?<br>3. Are you satisfied with who you are?<br>4. Do you feel safe in general?                                                                                                                                                                                                                                                                                                                            |
| Mood          | 1. Were you brave or scared?<br>2. Were you full of energies or weak?<br>3. Were you relaxed or worried?<br>4. Were you calm or couldn't help to stay seated?<br>5. Were you hungry or couldn't eat much?<br>6. Did you sleep well or not?<br>7. Were you happy or sad?<br>8. Were you peaceful or angry?<br>9. Were you physically ok or had stomach ache/headache?<br>10. Did you feel you were important or unworthy?<br>11. Did you feel surrounded by people who loved you or you felt alone? |

Table A2: Treatment impact on School aspirations

|               | Academic |        | Technical |        | Professional |        | No school |        | Don't know |        |
|---------------|----------|--------|-----------|--------|--------------|--------|-----------|--------|------------|--------|
|               | (1)      | (2)    | (3)       | (4)    | (5)          | (6)    | (7)       | (8)    | (9)        | (10)   |
| Treat         | 0.10     |        | -0.03     |        | -0.05        |        | -0.01     |        | -0.05      |        |
|               | (0.09)   |        | (0.39)    |        | (0.24)       |        | (0.85)    |        | (0.37)     |        |
|               | [0.17]   |        | [0.42]    |        | [0.32]       |        | [0.89]    |        | [0.41]     |        |
| ItalianXT     |          | 0.06   |           | -0.00  |              | -0.04  |           | -0.01  |            | -0.03  |
|               |          | (0.40) |           | (0.97) |              | (0.41) |           | (0.74) |            | (0.61) |
|               |          | [0.41] |           | [0.97] |              | [0.39] |           | [0.71] |            | [0.56] |
| ImmigrantXT   |          | 0.23   |           | -0.12  |              | -0.07  |           | 0.02   |            | -0.10  |
|               |          | (0.04) |           | (0.15) |              | (0.29) |           | (0.55) |            | (0.40) |
|               |          | [0.01] |           | [0.01] |              | [0.24] |           | [0.67] |            | [0.20] |
| Mean Control  | 0.43     |        | 0.09      |        | 0.14         |        | 0.06      |        | 0.27       |        |
| MC Italians   |          | 0.44   |           | 0.07   |              | 0.15   |           | 0.08   |            | 0.25   |
| MC Immigrants |          | 0.39   |           | 0.17   |              | 0.09   |           | 0.00   |            | 0.35   |
| R2            | 0.13     | 0.13   | 0.17      | 0.18   | 0.12         | 0.13   | 0.08      | 0.09   | 0.12       | 0.12   |
| Obs.          | 297      | 297    | 297       | 297    | 297          | 297    | 297       | 297    | 297        | 297    |

P-values from robust standard errors in parenthesis; p-values from randomized inference with 1000 repetitions in brackets. We control for gender, immigrant status, a dummy for missing information on immigrant status, single parent. Includes classroom fixed effects. Outcomes are dummies for self reported high school choice.

Table A3: Treatment impact on attitudes and wellbeing

|               | Grit   |        | Self-Efficacy |        | Wellbeing |        | Mood   |        | Index  |        |
|---------------|--------|--------|---------------|--------|-----------|--------|--------|--------|--------|--------|
|               | (1)    | (2)    | (3)           | (4)    | (5)       | (6)    | (7)    | (8)    | (9)    | (10)   |
| Treat         | -0.35  |        | 0.03          |        | -0.02     |        | -0.57  |        | -0.25  |        |
|               | (0.04) |        | (0.83)        |        | (0.91)    |        | (0.01) |        | (0.02) |        |
|               | [0.07] |        | [0.88]        |        | [0.94]    |        | [0.04] |        | [0.04] |        |
| ItalianXT     |        | -0.54  |               | 0.26   |           | 0.01   |        | -0.61  |        | -0.22  |
|               |        | (0.00) |               | (0.11) |           | (0.97) |        | (0.02) |        | (0.06) |
|               |        | [0.01] |               | [0.11] |           | [0.97] |        | [0.02] |        | [0.06] |
| ImmigrantXT   |        | 0.25   |               | -0.63  |           | -0.10  |        | -0.46  |        | -0.35  |
|               |        | (0.48) |               | (0.02) |           | (0.81) |        | (0.38) |        | (0.13) |
|               |        | [0.28] |               | [0.00] |           | [0.71] |        | [0.18] |        | [0.02] |
| Mean Control  | -0.00  |        | 0.00          |        | -0.00     |        | -0.00  |        | 0.01   |        |
| MC Italians   |        | 0.06   |               | -0.16  |           | 0.01   |        | 0.06   |        | -0.01  |
| MC Immigrants |        | -0.21  |               | 0.49   |           | -0.02  |        | -0.20  |        | 0.08   |
| R2            | 0.12   | 0.14   | 0.09          | 0.11   | 0.14      | 0.14   | 0.09   | 0.09   | 0.09   | 0.09   |
| Obs.          | 283    | 283    | 290           | 290    | 276       | 276    | 268    | 268    | 292    | 292    |

P-values from robust standard errors in parenthesis; p-values from randomized inference with 1000 repetitions in brackets. We control for gender, immigrant status, a dummy for missing information on immigrant status, single parent. Includes classroom fixed effects. Outcomes are standardized grit, self efficacy, wellbeing and mood measures constructed with Principal Component Analysis from items listed in Online Appendix 1, and a comprehensive index averaging the four.

Table A4: Treatment impact on School outcomes

|               | Italian |        | English |        | Maths  |        | GPA    |        | Absences |        |
|---------------|---------|--------|---------|--------|--------|--------|--------|--------|----------|--------|
|               | (1)     | (2)    | (3)     | (4)    | (5)    | (6)    | (7)    | (8)    | (9)      | (10)   |
| Treat         | 0.05    |        | -0.15   |        | 0.14   |        | -0.02  |        | 0.28     |        |
|               | (0.75)  |        | (0.31)  |        | (0.35) |        | (0.88) |        | (0.03)   |        |
|               | [0.78]  |        | [0.39]  |        | [0.44] |        | [0.90] |        | [0.05]   |        |
| ItalianXT     |         | 0.08   |         | -0.13  |        | 0.18   |        | 0.01   |          | 0.22   |
|               |         | (0.64) |         | (0.41) |        | (0.28) |        | (0.94) |          | (0.13) |
|               |         | [0.65] |         | [0.43] |        | [0.30] |        | [0.94] |          | [0.14] |
| ImmigrantXT   |         | -0.05  |         | -0.21  |        | -0.02  |        | -0.14  |          | 0.47   |
|               |         | (0.88) |         | (0.55) |        | (0.95) |        | (0.63) |          | (0.08) |
|               |         | [0.87] |         | [0.42] |        | [0.94] |        | [0.54] |          | [0.01] |
| Mean Control  | 0.00    |        | -0.00   |        | 0.00   |        | 0.00   |        | 0.00     |        |
| MC Italians   |         | -0.04  |         | -0.08  |        | -0.01  |        | -0.04  |          | 0.08   |
| MC Immigrants |         | 0.13   |         | 0.26   |        | 0.05   |        | 0.13   |          | -0.26  |
| R2            | 0.10    | 0.10   | 0.12    | 0.12   | 0.13   | 0.14   | 0.10   | 0.10   | 0.18     | 0.18   |
| Obs.          | 248     | 248    | 247     | 247    | 248    | 248    | 248    | 248    | 248      | 248    |

P-values from robust standard errors in parenthesis; p-values from randomized inference with 1000 repetitions in brackets. We control for gender, immigrant status, a dummy for missing information on immigrant status, single parent. Includes classroom fixed effects. Outcomes are standardized administrative grades in year 2021/22, GPA is computed by the researchers from administrative data as the average of all grades in year 2021/22, and absences refer to school year 2021/22.

## Online Appendix B

Below are the scripts of the video shown during the interviews.

- Control video, female actress: “We would like to thank you for agreeing to participate to the survey. Remember to answer in a sincere way and that you will not be judged based on your answers. Most of all, remember that you are doing something very important to the researchers. Many, many thanks.”
- Control video, male actor: “Hi! Thank you for agreeing to participate to the survey. Your answers are very important to the researchers, because thanks to those new activities and spaces for children will be created<sup>7</sup>, to help then learning while having fun. Remember: answer in a sincere way, as your answers will never be used to judge you nor they will be communicated to others, but only to know you better.”
- Treatment video, female actress: “Hi, I’m Alexandra, I’m 24-year-old and 12 years ago I moved to Italy. When I was at school, I did not feel enough compared to my classmates, I did not feel up to being able to do whatever I wanted to. It was probably due to the fact that some teachers kept saying to me that I would have not done anything good with my life. However, I remember that since when I was a child, I wanted to study art. Of course, I have never studied the right post-secondary school because the older ones kept saying to me that there was no job [in that field]. Lucky enough, I was brave enough to start the “Accademia di Belle Arti” [comparable to a university in figurative arts]. This was the best choice I could ever make. After three years I worked on my project facing several challenges, I am finally about to realize my first movie”.
- Treatment video, male actor: “Hi! Thank you for agreeing to participate to this survey.  
  
Today, I would like to tell you about my story. During school, for me it was not that

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<sup>7</sup>The survey is part of a wider intervention in collaboration with a local NGO, aimed at giving support to most fragile children in the sample.

easy, and seeing the sacrifices that my parents had to do every day was even harder. Moreover, I was not particularly talented for any subject and therefore I thought that after finishing the school, I would have not known what to do. What I learnt is that you have to understand what you really like and invest on your passions. Today, I am peaceful when I think about the results I achieved and I am happy about the path I walked through. Enjoy the rest of the survey!”

Figure A2 shows a screenshot taken from the two treatment videos; the same actors recorded also the control videos.

Figure A2: Screenshots from the two treatment videos

